

FORM TP 2015259



TEST CODE **02138020**

MAY/JUNE 2015

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. All working MUST be shown.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2014 Caribbean Examinations Council
All rights reserved.

02138020/CAPE 2015



0213802003

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (c) (i) Table 1 presents data representing the variation of the weight (W) of a satellite with its radial distance (R) from the Earth's centre as it orbits the Earth. On the grid provided in Figure 1 (page 5), plot a graph of weight (W) versus $1/R^2$ to reflect this variation. [4 marks]

TABLE 1: VARIATION OF WEIGHT OF SATELLITE WITH ITS RADIAL DISTANCE

$\frac{1}{R^2} \times 10^{-15} \text{ m}^{-2}$	Weight (W) $\times 10^5 \text{ N}$
21	1.18
18	1.05
16	0.95
12	0.66
9	0.58
7	0.40
6	0.35
4	0.24

- (ii) Find the gradient of the graph and hence determine the mass of the satellite. (Mass of Earth, $M_E = 5.6 \times 10^{24} \text{ kg}$)

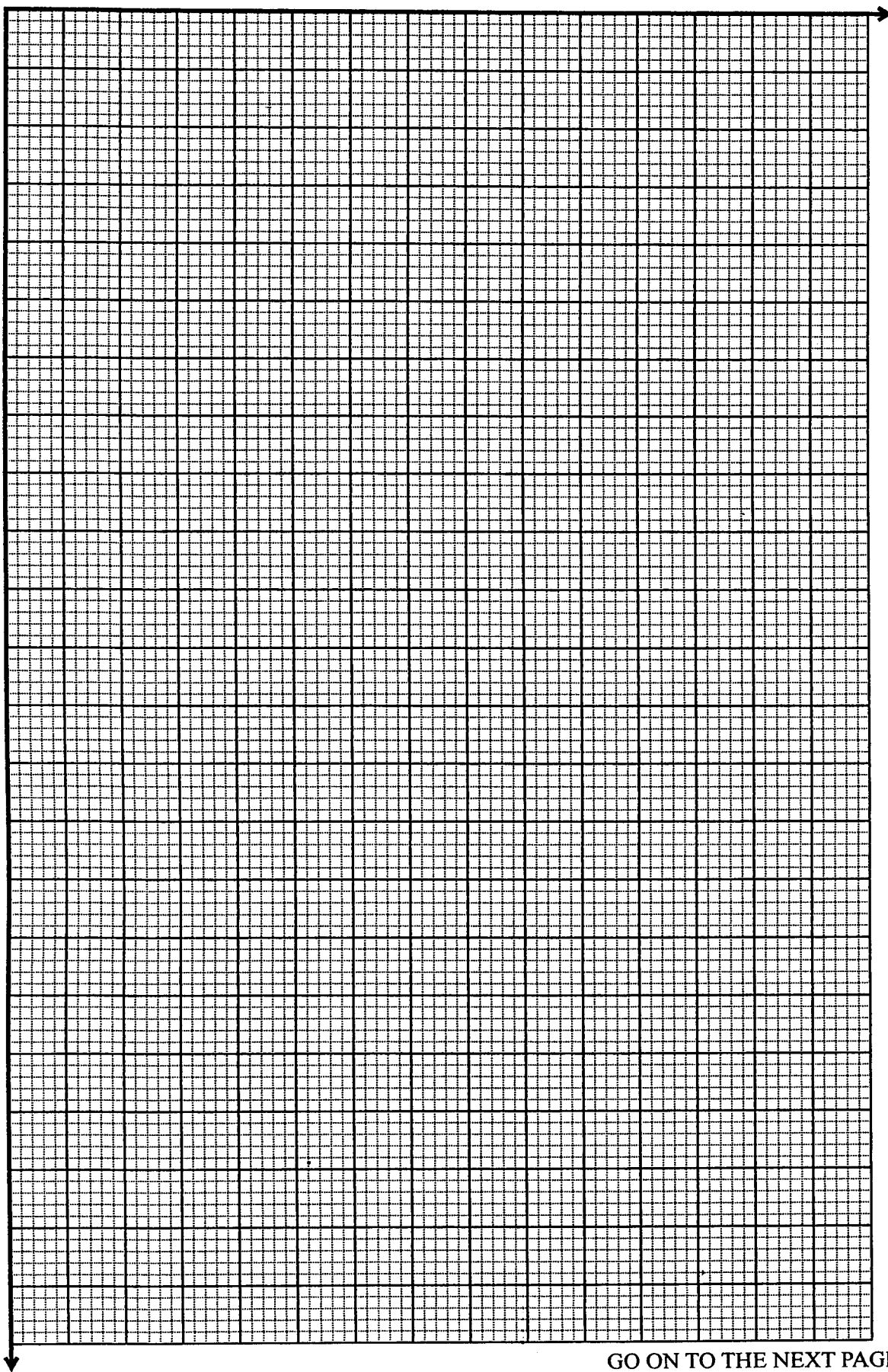
[4 marks]

Total 15 marks

GO ON TO THE NEXT PAGE



Figure 1. Variation of weight with radial distance



GO ON TO THE NEXT PAGE



2. (a) (i) State TWO conditions that must be satisfied for simple harmonic motion to occur.

.....

.....

.....

.....

[2 marks]

- (ii) A horizontal plate is vibrating vertically, as shown in Figure 2.

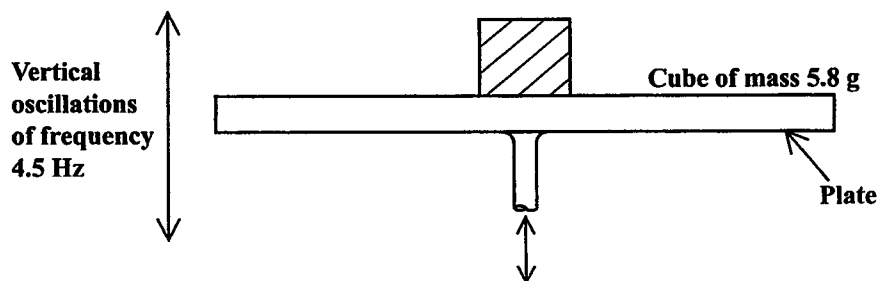


Figure 2. Horizontal plate

The plate undergoes simple harmonic motion with a frequency of 4.5 Hz and amplitude 3.0 mm. A metal cube of mass 5.8 g rests on the plate. Calculate the maximum kinetic energy of the cube.

.....

.....

.....

.....

.....

.....

[3 marks]

GO ON TO THE NEXT PAGE



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (iii) The amplitude of oscillation of the plate in (a) (ii) is gradually increased. The frequency remains constant. At a particular amplitude and at the position of maximum displacement, the cube loses contact with the plate for an instant. Explain why the cube loses contact with the plate.

.....

.....

.....

.....

.....

[2 marks]

- (iv) Calculate the amplitude of oscillation at the instant the cube loses contact with the plate.

[3 marks]



- (b) Figure 3 shows a block which oscillates horizontally between two identical springs, each with spring constant, k . The vibrator has a constant amplitude.

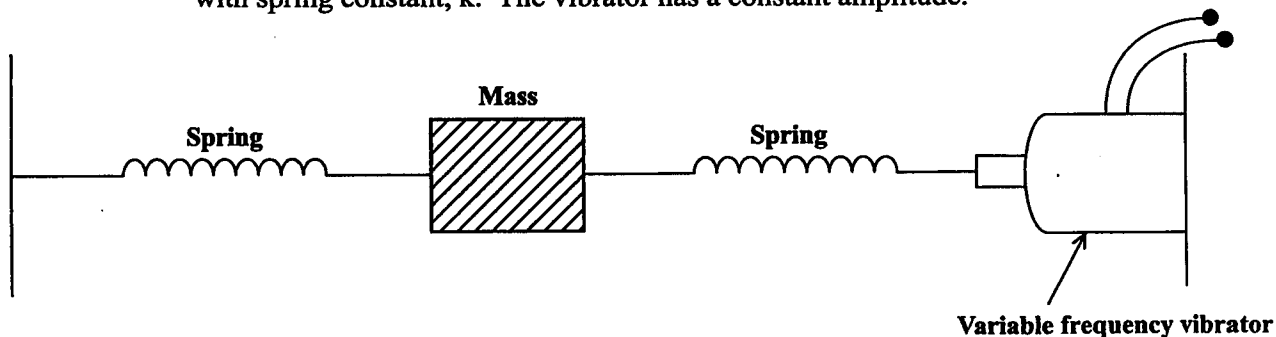


Figure 3. Oscillating block

As the frequency is changed, the amplitude of oscillation of the block is measured. Table 2 shows the data recorded.

TABLE 2: VARIATION OF AMPLITUDE WITH FREQUENCY

Frequency f/Hz	Amplitude y/cm
2.8	1.2
5.5	1.8
8.3	3.0
11.0	4.2
12.7	4.4
13.8	4.3
16.5	1.4

- (i) On the grid provided in Figure 4 (page 9), plot a graph of amplitude of oscillation, y , against frequency, f . [4 marks]
- (ii) Use your graph to determine the angular frequency for the block when it is oscillating at maximum amplitude.

Angular frequency

[1 mark]

Total 15 marks

GO ON TO THE NEXT PAGE



DO NOT WRITE IN THIS AREA

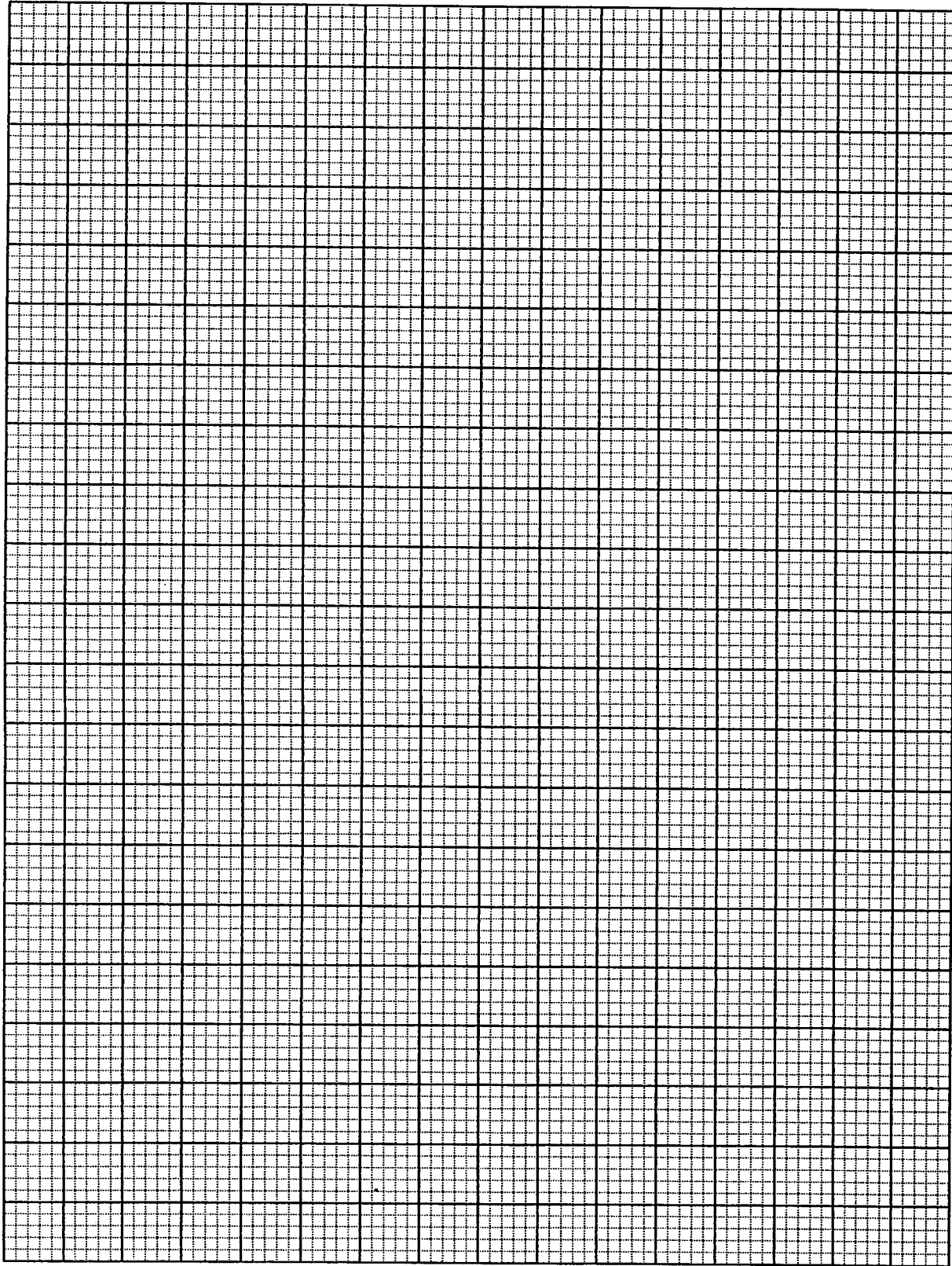


Figure 4. Amplitude of oscillation versus frequency



3. (a) A resistance thermometer and a thermocouple thermometer are both used at the same time to measure the temperature of a water bath. Explain why the thermometers may record different temperatures even though they are calibrated accurately and are at equilibrium.

.....

.....

.....

.....

[2 marks]

- (b) (i) What is meant by the term 'absolute zero of temperature' ?

.....

.....

[1 mark]

- (ii) State how the 'absolute thermodynamic scale of temperature' differs from other temperature scales.

.....

.....

.....

[1 mark]

- (c) The temperature of a water bath increases from 50.00 °C to 80.00 °C. Determine, in kelvin, and to an appropriate number of significant figures,

- (i) the temperature 50.00 °C

[1 mark]

- (ii) the change in temperature of the water bath.

[1 mark]

GO ON TO THE NEXT PAGE



- (d) An empirical centigrade temperature can be determined and defined in terms of any suitable thermometric property. Select ONE type of thermometer with a suitable thermometric property and describe briefly the procedure and necessary readings to be taken in order to use this thermometer to determine a random temperature, θ .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5 marks]

- (e) The relationship between the resistance, R , of a wire at temperature $t/^{\circ}\text{C}$ as measured by a mercury thermometer, and the resistance, R_0 , at 0°C is

$$R = \frac{R_0}{1 + \alpha t^2},$$

where $\alpha = 5.0 \times 10^{-5} \text{ }^{\circ}\text{C}^{-2}$.

- (i) Calculate, in terms of R_0 , the resistance of the wire at 100°C and at 80°C as measured by the mercury thermometer.

[2 marks]

GO ON TO THE NEXT PAGE



- (ii) If the resistance of the wire is used as a thermometric property, what is the temperature reading of the resistance thermometer when the reading on the mercury thermometer is 80°C ?

[2 marks]

Total 15 marks

DO NOT WRITE IN THIS AREA



SECTION B

Answer ALL questions.

Write your answers in the spaces provided.

4. (a) Figure 5 shows a resultant vector, A , at an angle θ to the horizontal.

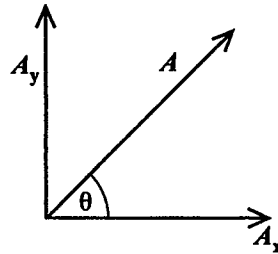


Figure 5. Resultant vector

Write expressions for A_x and A_y in terms of A and θ .

.....

.....

.....

[2 marks]

- (b) A jogger runs 145 m in a direction 20.0° east of north and then 105 m in a direction 35.0° south of east.

- (i) Represent his resultant displacement using a suitable vector diagram.

[3 marks]

GO ON TO THE NEXT PAGE



- (ii) Determine his northerly displacement from his starting point.

[3 marks]

- (iii) Hence, determine his resultant displacement from his starting point.

[4 marks]

GO ON TO THE NEXT PAGE



- (c) It is said that sprinters in a 100 m race accelerate for the first one-third of the race. The following formula, $t = \left(\frac{2s}{a} \right)^x$ is proposed to compute the duration of the acceleration phase. By comparing the units, find the value of x .

[3 marks]

Total 15 marks

GO ON TO THE NEXT PAGE



5. (a) Long-sightedness is an eye defect that affects many persons. Explain, using suitable optical sketches, how long-sightedness occurs and how it can be corrected.

[4 marks]

- (b) Use a ray diagram to show how focused images are formed in a magnifying glass.

[3 marks]

GO ON TO THE NEXT PAGE



- (c) An object is placed 12 cm from a converging lens of focal length 18 cm.

Determine

- (i) the position of the image and its distance from the lens

[4 marks]

- (ii) whether the image is virtual or real

[1 mark]

- (iii) the magnification of the image and hence its orientation.

[3 marks]

Total 15 marks

GO ON TO THE NEXT PAGE



DO NOT WRITE IN THIS AREA

6. (a) Define the term 'Young's modulus'. Include a suitable formula in your definition.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[3 marks]



- (c) A bone has a Young's modulus of approximately 1.8×10^{10} Pa. Under compression, it can withstand a stress of about 1.5×10^8 Pa before breaking. Assuming that the femur (thigh bone) is 0.47 m long, calculate the amount of compression that this bone can withstand before breaking. (State your answer in mm.)

[3 marks]



- (d) Figure 6 shows a crane that is used for lifting heavy objects.

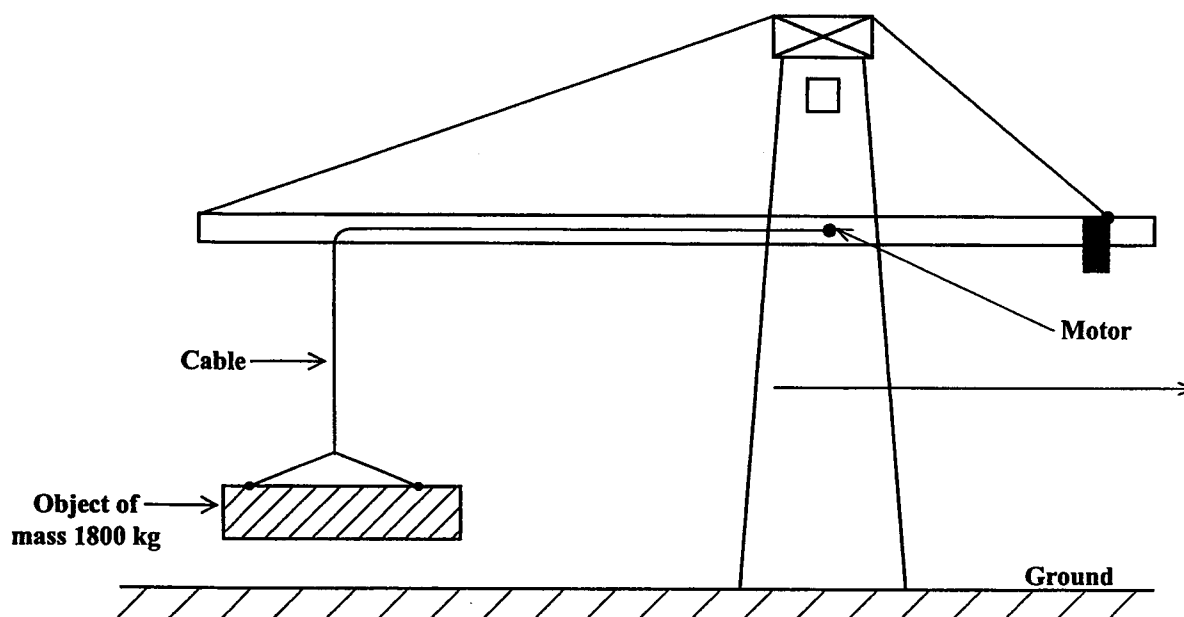


Figure 6. Crane

The motor in the crane lifts a total mass of 1800 kg from rest on the ground. The cable supporting the mass is made from steel of Young's modulus 2.4×10^{11} Pa. The cross-sectional area of the cable is $1.3 \times 10^{-4} \text{ m}^2$. As the mass leaves the ground, the strain in the cable is 0.0010.

(Assume the weight of the cable is negligible.)

- (i) Use the Young's modulus of the steel to determine the tension in the cable.

[2 marks]

GO ON TO THE NEXT PAGE



- (ii) Assuming it is an 8-tonne crane and that the length of the cable is 8 m, determine the extension in the cable when it is operating at maximum load.

[3 marks]

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

02138020/CAPE 2015



0213802025

EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



0213802030

02138020/CAPE 2015